

AI Logistics Simulator User Manual

This manual lists every control and every output metric, with **allowed values / ranges, units**, and **how to read** them in the prototype.

1) Before You Run

1. Start the FastAPI backend: `uvicorn backend.app:app --reload`
2. Start the simulator UI: `streamlit run backend/streamlit_control_tower.py`
3. Open the Streamlit URL from the terminal (commonly `http://localhost:8501`).

After you change backend logic, **restart the API** (or rely on `--reload`); the UI only needs a refresh if the page is stale.

2) Sidebar Controls

Control	Range / values	Default	Interpretation
FastAPI Base URL	Any valid HTTP base URL for your API	<code>http://127.0.0.1:8000</code>	Where the UI sends <code>POST</code> requests. Must match the running <code>uvicorn</code> host and port.
Mode	<code>sync</code> · <code>async</code> · <code>replay</code>	<code>sync</code>	sync : standard orchestration. async : same flow via async runtime. replay : same result plus a per-node execution trace (timeline).
View	<code>Classic Simulator</code> · <code>10 Challenges Demo</code>	<code>10 Challenges Demo</code>	Classic : summary KPIs + notes + replay trace. 10 Challenges : one tab per use case (demand, route, ...) with detail.
Show existing raw JSON output	On / Off	Off	On : shows raw <code>execution_meta</code> and <code>tool_outputs</code> for debugging. Off : business-friendly view only.

3) Scenario Form (Inputs)

All of these are sent in the JSON body when you click **Run Simulation**. Sliders use **inclusive** min/max as shown in the UI.

Field	Type	Range / allowed values	Default	Step	Interpretation & effect in the prototype
Region	Select	<code>SG</code> · <code>MY</code> · <code>TH</code> · <code>PH</code> · <code>ID</code> · <code>VN</code> · <code>HK</code> · <code>TW</code>	<code>SG</code>	—	Label for the forecast service; does not change numeric heuristics unless you extend the model.
Demand Index	Slider	<code>0.60</code> – <code>1.80</code> (unitless index)	<code>1.15</code>	<code>0.05</code>	1.0 = baseline demand . Below 1.0 reduces pricing pressure; above 1.0 increases it (see Demand multiplier under outputs). Also nudges last-mile failure slightly when > 1.0.
Weather	Select	<code>clear</code> · <code>rain</code> · <code>storm</code>	<code>clear</code>	—	Affects route ETA (rain/storm = longer ETA) and last-mile failure (mapped to internal severity for the failure model).
Urgency Mix	Select		<code>express</code>	—	Affects dynamic price (higher urgency → higher multiplier) and carbon

Field	Type	Range / allowed values	Default	Step	Interpretation & effect in the prototype
		standard · express · same_day			recommendation text; nudges last-mile failure (tighter service = slightly higher risk in the prototype).
Fleet Health Index	Slider	0.50 – 1.00 (0 = poor, 1 = excellent)	0.90	0.01	Drives simulated prior failed delivery attempts (lower health → more “prior attempts” in the failure model) and signals maintenance context in the story; maintenance tool still uses its own fixed demo inputs unless you extend the API.
Traffic Level	Select	low · medium · high	medium	—	Affects route ETA and fuel (higher traffic → longer ETA, more fuel) and nudges customer presence in the last-mile failure model.
Vehicle Type	Select	diesel_van · ev_van · hybrid_truck	diesel_van	—	Affects route fuel and CO₂ from the route tool ; carbon tab uses a separate per-vehicle emissions factor.
Tool Retries	Integer slider	0 – 3 (count)	1	1	How many retries each tool call may use after a failure/timeout. Higher = more resilience, longer worst-case wait.
Tool Timeout (s)	Slider	1.0 – 10.0 seconds	3.0	0.5	Max time per tool execution. Lower = fail fast; higher = allow slow steps.

4) How to Run a Scenario

1. Set **Mode**, **View**, and optional **raw JSON** in the sidebar.
2. Adjust the scenario form.
3. Click **Run Simulation**.
4. Read the **status** line, then the **metrics** for your selected view (see sections 5–6).

5) Top dashboard metrics (Plan + classic header)

These come from the orchestration **plan** and **execution_meta** and appear in **Classic** view and/or the **10 Challenges** header row.

Metric	Typical range in prototype	Unit	Where it comes from	How to read it
Target ETA (min)	About tens of minutes (depends on traffic × weather × fixed route geometry in code)	Minutes	Route tool: distance + traffic penalty × weather penalty → <code>estimated_eta_min</code>	Lower is faster. Rises with high traffic and rain/storm . Not tied to Region text.
Price Quote	≥ 3.50 in the pricing heuristic	Currency units	Pricing tool: base distance + weight + urgency multiplier × demand multiplier + fuel surcharge	Higher urgency / distance / demand → higher quote.

Metric	Typical range in prototype	Unit	Where it comes from	How to read it
	(then scaled by urgency and demand)	(unnamed in demo)		Demand multiplier = $1 + (\text{demand_index} - 1) \times 0.2$ (so at index 1.0 it is 1.0).
Failure Probability	0.000 – 0.990 (capped)	Probability (0–1)	Last-mile / delivery failure tool: weighted mix of customer presence , address quality , weather severity , prior attempts , COD	Higher = more last-mile risk in the toy model. Varies with weather , traffic , fleet health , demand , urgency (see section 3).
Forecast Volume	Non-negative integer (shipment count style)	Count / day (conceptual)	Forecast tool: rolling history + promo/holiday flags (defaults are fixed in orchestration)	Higher volume vs baseline tends to higher capacity risk.
CO₂ Estimate (kg) (10 Challenges header)	≥ 0 , scales with route distance and fuel heuristic	kg CO ₂ (approximate)	Route metrics: <code>co2_kg_estimate</code> $\approx$ <code>fuel_liters</code> $\times$ 2.68	Lower is better for the same trip; vehicle and traffic change fuel in the route model.
Dispatch Buffer (%)	3, 8, or 15 in the current forecast rules	Percent	Forecast <code>recommended_buffer_pct</code> from capacity_risk	Higher buffer when capacity_risk is high (suggested extra operational slack).
Risk Review Triggered	Yes / No	Boolean	<code>execution_meta.risk_review_triggered</code> (tied to whether <code>"risk_review"</code> appears in agent notes in the current code)	Yes = risk path was noted in the trace; use replay + raw JSON to inspect.
Trace Steps (replay mode)	Integer ≥ 1	Count	Replay stream: one step per graph event	Larger = more node updates recorded in the trace.
Agent Notes	List of short strings	—	Appended by each agent node in the graph	Narrative of what the orchestration “did” in order.
Replay Timeline (replay mode)	Steps with node name , elapsed ms , keys updated	Time (ms)	Replay trace	Explainability : which node ran and which state keys changed.

6) “10 Challenges Demo” tab metrics

Each tab shows tool output. Ranges below match the **prototype** math in the backend, not live production data.

Tab 1 – Demand (forecast)

Metric	Range / values	Unit	Interpretation
Predicted Volume	Non-negative integer	Count	Model “next day” style volume from historical series + promo/holiday multipliers (defaults fixed unless scenario extended).
Trend	upward · stable · downward	—	Class vs recent baseline ($\pm 15\%$ / -10% style thresholds in code).
Capacity Risk	low · medium · high	—	high if volume > baseline $\times 1.2$; medium if above baseline; else low .
Recommended buffer %	3 / 8 / 15	%	Tied to capacity_risk (high $\rightarrow$ 15, medium $\rightarrow$ 8, low $\rightarrow$ 3).

Tab 2 – Route (optimize_route)

Metric	Range / values	Unit	Interpretation
ETA (min)	Positive integer (prototype heuristic)	Minutes	Same as Target ETA drivers: traffic $\times$ weather $\times$ fixed distance/waypoints.
Distance (km)	$\geq \sim 20$ with default waypoints	km	Base distance from segment length + waypoints (hard-coded list length in orchestrator).
Fuel (L)	Positive decimal	Liters	Heuristic from distance / efficiency $\times$ traffic; not a calibrated fleet figure.
Route	Ordered list of strings	—	Origin $\rightarrow$ waypoints $\rightarrow$ destination path labels.

Tab 3 – Warehouse

Metric	Range / values	Unit	Interpretation
Estimated Walk Reduction	≤ 35 (cap), grows with task count in prototype	%	Illustrative pick-path savings vs naive ordering.
High-demand zones	List of aisle labels	—	Zones where demand_score ≥ 0.8 on demo tasks.
Optimized sequence	Ordered steps	—	Pick tour order for the demo SKU list.

Tab 4 – Maintenance

Metric	Range / values	Unit	Interpretation
Risk	Low · Medium · High	—	From risk_score thresholds: ≥ 0.7 High, ≥ 0.45 Medium, else Low (demo inputs fixed in orchestrator).
Risk Score	0 – $\sim 1+$ before clamping-style bands	Score	Weighted blend of temperature , vibration , mileage , DTC count (prototype weights).
Action	Text	—	Recommended next step for that score band.

Tab 5 – Fraud

Metric	Range / values	Unit	Interpretation
Fraud Probability	0.000 – 0.990	Probability	Blended score from claim size , claim frequency , missing docs , account age (fixed demo claim in orchestration).
Decision	auto_process · manual_review	—	manual_review if fraud probability ≥ 0.6 .

Tab 6 – Customer Service (chatbot)

Metric	Range / values	Unit	Interpretation
Answer	Text	—	Rule-based template from query / tracking_id .
Escalate to human	Yes / No	—	Yes if top retrieval confidence below threshold or query contains complaint (prototype rules).

Tab 7 – Pricing

Metric	Range / values	Unit	Interpretation
Recommended Price	≥ 3.5 after rounding	Currency (demo)	Final dynamic price .
Explainability	Base, urgency ×, demand ×, fuel surcharge	—	Urgency : standard 1.0 , express 1.25 , same_day 1.55 . Demand multiplier scales with Demand Index around 1.0.

Tab 8 – Last-Mile Failure

Metric	Range / values	Unit	Interpretation
Failure Probability	0.000 – 0.990	Probability	Same model as dashboard Failure Probability (see section 5).
Action	normal_dispatch · confirm_delivery_window	—	confirm_delivery_window if failure probability ≥ 0.5 .

Tab 9 – Carbon

Metric	Range / values	Unit	Interpretation
Estimated Emissions (kg)	≥ 0	kg	Distance × per-vehicle_type factor (EV uses grid_carbon_intensity scaling).
Saving vs Diesel	0 – 100 (percentage points)	%	Improvement vs a diesel baseline for the same distance.
Recommendation	switch_to_ev · optimize_consolidation	—	switch_to_ev when urgency is standard and distance ≤ 120 km in the prototype; otherwise consolidation hint.

Tab 10 – Multi-Agent (summary)

Repeats **Dispatch Buffer**, **Final Price**, **Failure Probability**, **Agent Notes**, and optionally **Replay Trace** — same interpretation as in sections 5–6.

7) Recommended Usage Patterns

- **A/B testing:** Fix all inputs except one (e.g. only **Weather**) and compare **ETA**, **Price**, **Failure Probability**, **CO₂**.
 - **Reliability:** Sweep **Tool Retries** (0 → 3) and observe errors/timeouts in **raw JSON**.
 - **Latency stress:** Lower **Tool Timeout** toward **1.0 s** and combine with high **Traffic** / **storm**.
 - **Explainability:** **Mode** = **replay** + **Show raw JSON** to align numbers with **trace** steps.
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8) Troubleshooting

Issue	What to check
Connection errors	FastAPI Base URL matches <code>uvicorn</code> ; firewall/host binding.
HTTP errors	Enable raw JSON ; read response body from API.
Empty replay trace	Mode must be replay ; backend must expose <code>/agents/control/replay</code> .
Metrics unchanged after code edits	Restart backend (or confirm <code>--reload</code> picked up files).

For onboarding, **change one input at a time** and compare the **plan** metrics (section 5) before diving into every challenge tab.